

Six-Recommendation Integration: Cross-Omic Decoupling of the Spaceflight Kidney DCT/NCC Phenotype

RRRM-2 Kidney Transcriptome project — consolidated results addendum

2026-05-22

This addendum reports the six analytical recommendations of the May 2026 strategy memo, run end-to-end and integrated into the v10 phenotype-anchored frame. The guiding thesis is unchanged and is the one the memo endorses: *the established spaceflight DCT/NCC phenotype sits inside a recurrent kidney-wide RNA remodeling context that is cross-omically decoupled — it recurs in transcriptomes, fails as a protein-abundance story, and resolves at the NCC/SPAK regulatory-phosphorylation layer*. Each recommendation is graded conservatively. Numbers are drawn from the run artifacts `run_20260522_osd462_anchor`, `run_20260522_regulator_activity`, `run_20260522_phenotype_anchor`, `run_20260522_celltype_decomposition`, and `run_20260522_physiology_assessment`.

Recommendation 3 — Cross-omic transcript→protein→phospho dissociation [supported as the central result]

The OSD-462 / RR-10 anchor resolves the memo’s central claim. (i) **RNA recurrence**: the matrix-high/DCT-low pathway vector recurs between RRRM-2 ISS-T and OSD-462 RNA (cosine = 0.869, bootstrap CI [0.65, 0.90], leave-one-pathway-out range 0.82–0.90; 11 shared pathways) — **gate PASS**. (ii) **Protein abundance does not mirror RNA**: the genome-wide RNA↔protein flight-effect Spearman is 0.043; the ECM family signed-mean protein effect is -0.107 (*opposite* to the RNA-predicted increase; cross-study concordance -0.40) and the DCT family is null (-0.044); no targeted set is concordant in its predicted direction (**H1, H2 falsified**). (iii) **Phospho layer**: total NCC protein is flat ($+0.089$), yet the NCC N-terminal activating cluster (mean -0.83) and SPAK/OSR1 sites (mean -0.41) are strongly suppressed — 7 NCC-regulatory and 3 SPAK/OSR1 sites significant-down, non-regulatory *Slc12a3* sites flat (**H3 supported**). (iv) **Network translation**: RRRM-2 network-candidate genes are *not* enriched among protein- or phospho-changing genes ($p_{\min} = 0.23/0.25$; **H4 falsified**). The transporter lesion is a post-translational/activity event embedded in an RNA context, not a protein-abundance signature.

Recommendation 1 — Frozen RNA predictor / phenotype anchor [group supported; per-animal underpowered]

A phosphoproteomics-naïve DCT/NCC-WNK transport RNA score and a pre-specified NCC regulatory-phospho activity score (*Slc12a3* T53/58/65/68, *Stk39* S382/383) were matched across 20 RR-10 animals. **Group level (concordant)**: flight lowers both the RNA program (-0.324) and NCC regulatory phosphorylation (-1.298); the regulatory cluster is suppressed $\sim 4.3\times$ more than non-regulatory NCC sites (-1.30 vs -0.30). **Per animal**: the link is positive but not significant (Spearman $+0.328$, $p = 0.16$; condition-adjusted $+0.365$, $p = 0.11$) — the anticipated underpowered-at- $n=20$ outcome. **Controls**: dropping the *Slc12a3* transcript preserves the group concordance (-0.247); but non-regulatory NCC phosphosites *also* covary per-animal with the RNA score ($+0.555$, $p = 0.011$), so per-animal specificity is *not* exclusive to

Cross-omic decoupling of the spaceflight kidney DCT/NCC phenotype — six-recommendation integration

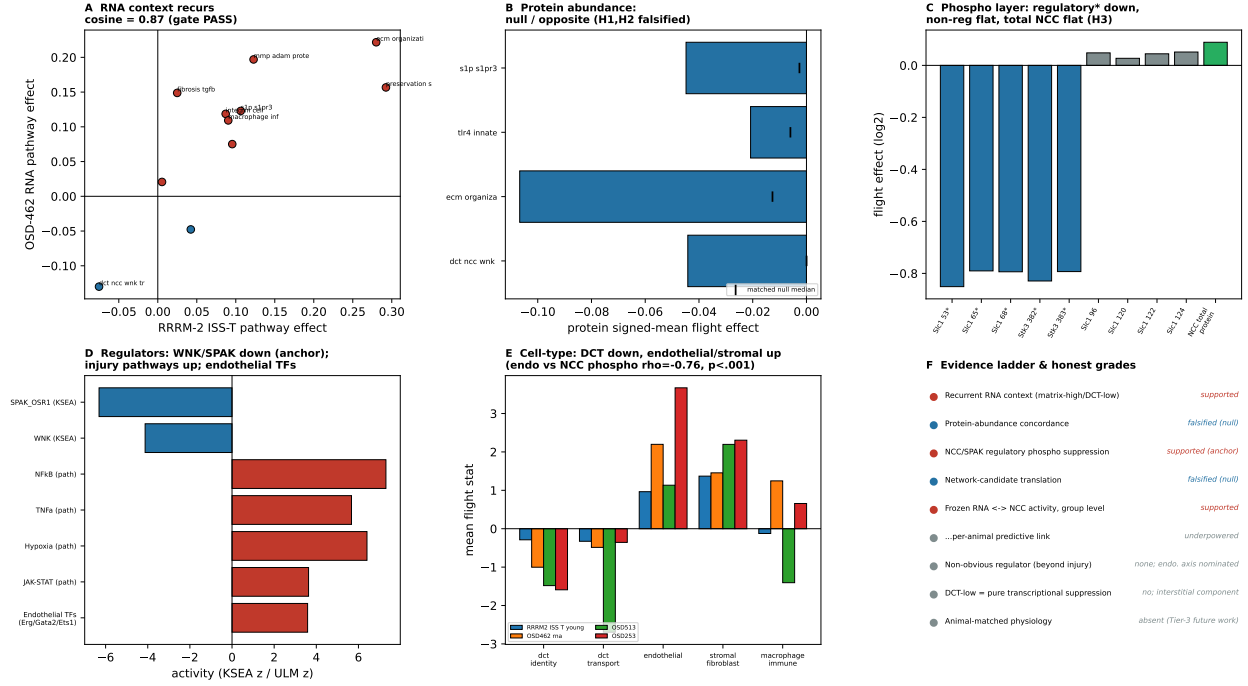


Figure 1: **Consolidated six-recommendation integration.** (A) The matrix-high/DCT-low RNA pathway vector recurs between RRRM-2 ISS-T and OSD-462 (cosine = 0.87, gate PASS). (B) Matched-null protein-abundance concordance is null/opposite for both matrix (ECM) and DCT families (H1, H2 falsified). (C) On the WNK–SPAK/OSR1–NCC chain, activating regulatory phosphosites (asterisk) are strongly suppressed while non-regulatory *Slc12a3* sites and total NCC protein are flat (H3 supported). (D) KSEA returns WNK and SPAK/OSR1 inferred-activity down (anchor); decoupler recurrent pathway activity is dominated by expected injury programs, with an endothelial/vascular TF module as the most coherent non-obvious recurrent signal. (E) Marker-panel decomposition: DCT identity *and* transport fall while endothelial/stromal compartments rise across cohorts; per animal the endothelial score anti-correlates with NCC regulatory phospho ($\rho = -0.76$, $p < 0.001$). (F) Evidence ladder and honest grades.

regulatory sites. The honest claim ceiling is therefore group-level concordance with magnitude specificity, not a validated per-animal predictor; reported as such.

Recommendation 2 — Search for a non-obvious regulator [KSEA anchor **supported**; Layer B nomination]

Layer A (KSEA, OSD-462 phosphoproteome): the WNK–SPAK/OSR1 positive control passes — SPAK/OSR1 $z = -6.31$ ($p = 2.8 \times 10^{-10}$), WNK $z = -4.12$ ($p = 3.7 \times 10^{-5}$), both inferred-activity-down — certifying the activity-inference machinery. **Layer B (decoupler ULM; PROGENy + CollecTRI, mouse):** run within each of five cohorts (RRRM-2 ISS-T, RRRM-2 LAR, OSD-462, OSD-513, OSD-253), never by pooling. Across the four terminal-flight cohorts the recurrently-up pathways are the *expected* injury/inflammation programs (NF- κ B, TNF α , JAK-STAT, Hypoxia) — all of which *reverse sign* in the RRRM-2 LAR live-return arm, consistent with recovery. Of 751 scored TFs, none recur as inferred-down; the most coherent non-obvious recurrent signal is an **endothelial/vascular TF module** (*Erg*, *Gata2*, *Ets1*), consistent with the S1P/S1PR3 and adhesion components of the RNA neighborhood, with *Sp1* (macrophage) and retinoid receptors (*Rara/Rarg*) also recurrent. **Verdict (claim discipline):** no single non-obvious upstream driver dominates beyond known injury/stress pathways; the endothelial/vascular axis

is nominated as a candidate context axis, not a proven regulator.

Recommendation 4 — Cell-type / spatial evidence [bounds interpretation]

A curated marker-panel decomposition separates DCT *identity* (*Pvalb*, *Trpm6*, *Calb1*, ...) from the DCT *transport* program. Across the four flight cohorts, *both* DCT identity (OSD-462 -1.00 , OSD-513 -1.48 , OSD-253 -1.59) and DCT transport fall, while endothelial (OSD-462 $+2.20$) and stromal/fibroblast (OSD-462 $+1.45$) compartments rise recurrently. In OSD-462 the per-sample endothelial ($+1.00$, $p = 0.036$) and stromal ($+0.67$, $p = 0.054$) flight increases are the strongest moves, and **per animal the endothelial score anti-correlates with NCC regulatory phospho** ($\rho = -0.76$, $p = 0.0004$); stromal likewise ($\rho = -0.51$, $p = 0.023$). Because DCT identity falls alongside transport and interstitial compartments expand, the bulk DCT-low signal carries a genuine composition / interstitial-expansion component and *cannot* be read as pure cell-autonomous transcriptional suppression. This is precisely why the regulatory-phosphosite layer — normalized within the phosphoproteome and $\sim 4\times$ more suppressed at regulatory than non-regulatory sites — is the more reliable transporter readout. Cell-type analysis here *bounds* interpretation and reinforces the cross-omic decoupling thesis rather than overturning it.

Recommendation 5 — Physiology [absent in-cohort; Tier-2/3 literature ladder]

The OSD-462 record contains only molecular assays; no urine/serum electrolyte, blood-pressure, or per-animal kidney-function endpoint is present, so no animal-level physiology $\leftrightarrow$ RNA inference is possible. The same-mission Cosmic Kidney Disease study (Siew et al. 2024) supplies a **Tier-2 morphology anchor**: flight reduces DCT NCC-positive (tNCC) density by 47.6% ($p = 0.001$; total area -26.3% , $p = 0.082$; mean object area $+34.1\%$, $p = 0.028$), a morphological correlate consistent with the measured NCC/SPAK phospho suppression. Group-level plasma and urine electrolytes (Na, Cl, Ca, Mg, K, PO_4) and fractional-excretion readouts exist in that study but cannot be matched to the OSD-462 omics animals. Per the memo: state the absence explicitly; animal-matched electrolyte/blood-pressure endpoints remain Tier-3 future work.

Recommendation 6 — Kill weak branches (main vs supplement triage)

Table 1 is the placement decision derived from the current evidence grades. The ruthless main-text spine is: (1) prior work established the DCT/NCC phenotype; (2) a DCT/NCC-low RNA context recurs across RRRM-2, OSD-513, OSD-462; (3) that context is matrix/adhesion/proteolysis/S1P/innate/stress-up, DCT/NCC-WNK-down; (4) protein abundance does not mirror it; (5) NCC/SPAK phosphoproteomics + KSEA locate the transporter phenotype at the activity layer; (6) animal-matched analysis supports group-level concordance, not per-animal prediction; (7) marker panels show an interstitial-expansion component bounding the "state" language; (8) regulator, network, age, and LAR analyses define boundaries and a single nominated endothelial context axis.

Consolidated evidence grades

What this paper claims / does not claim. It contributes a reproducible, cross-omically decoupled RNA context map around an *established* phenotype, with explicit null boundaries (protein abundance, network translation), an interstitial-expansion caveat on the bulk DCT signal, and a single nominated endothelial/vascular context axis. It does *not* discover NCC dephosphorylation (Siew et al. 2024), prove a causal driver, claim a validated per-animal predictor, or establish physiology without direct endpoints.

Component	Placement	Grade / reason
Cross-cohort RNA recurrence (matrix-high/DCT-low)	main	supported (cosine 0.87)
OSD-462 protein-abundance concordance	main	falsified — the punchline
NCC/SPAK regulatory phospho + KSEA	main	supported (Tier-1 anchor)
Phenotype-anchor (animal-matched RNA↔NCC)	main (careful)	group supported , per-animal underpowered
Cell-type marker-panel decomposition	main (short)	bounds interpretation
Regulator Layer B (TF/pathway recurrence)	main-short / supp.	nomination (endothelial axis)
Physiology ladder (Siew tNCC morphometry)	discussion	Tier-2 literature anchor
Network-candidate translation (Layer 3)	supplement	falsified (neg. method)
Age-rewiring failure	supplement + boundary	not supported (history)
LAR attenuation/divergence	supplement	context (recovery reversal)
OSD-253 / TLR4	supplement	association, not pillar
WGCNA grey60	supplement	annotation
LIONESS / node2vec	supplement	falsified (neg. method)
External aging projection	supplement	rules out aging mimicry

Table 1: Main-text vs supplement triage (memo Rec 6), graded from the current results.

Claim	Grade
Matrix-high/DCT-low RNA context recurs across cohorts	supported
RNA recurrence translates to matrix/transporter protein abundance	falsified
NCC/SPAK regulatory-phosphorylation suppression (activity layer)	supported (anchor)
Network-candidate genes translate to protein/phospho changes	falsified
Frozen RNA score tracks NCC activity — group level	supported
Frozen RNA score predicts NCC activity — per animal	underpowered
A non-obvious upstream regulator beyond injury programs	nomination (endothelial axis only)
Bulk DCT-low is pure DCT transcriptional suppression	no (interstitial component)
Animal-matched physiology endpoint	absent (Tier-3 future work)

Table 2: Honest claim grades after running all six recommendations.